Prefect Approach on Fugaku

- Running SBD as an HPC task from Python
- Building hybrid quantum-classical workflows on Fugaku

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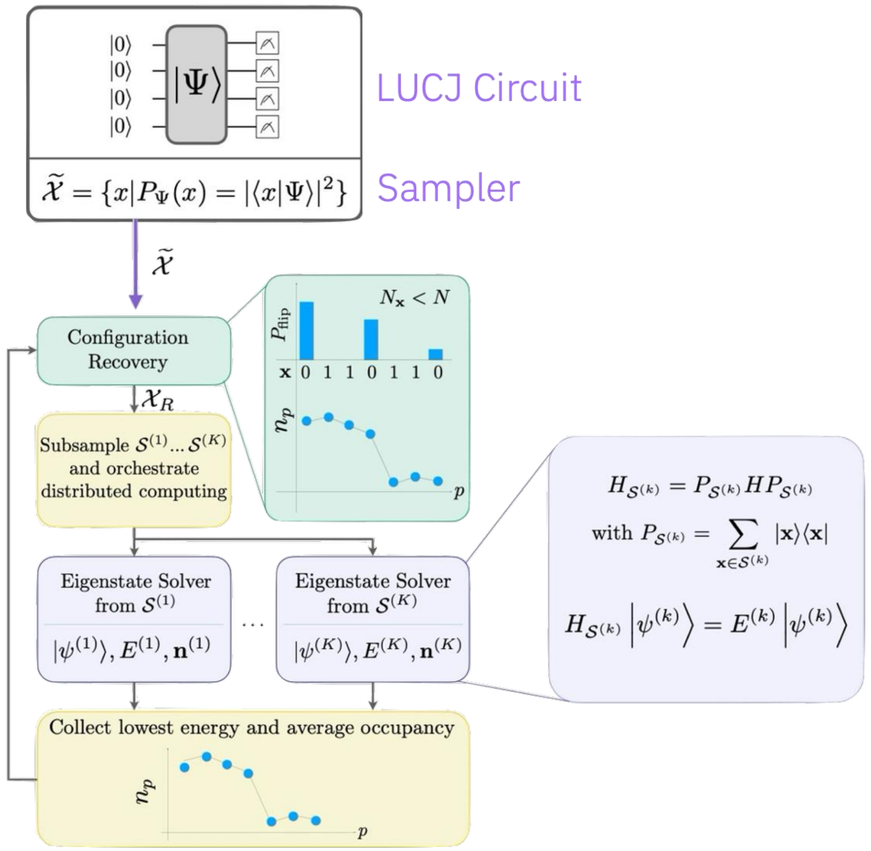
Why Workflow Orchestration?

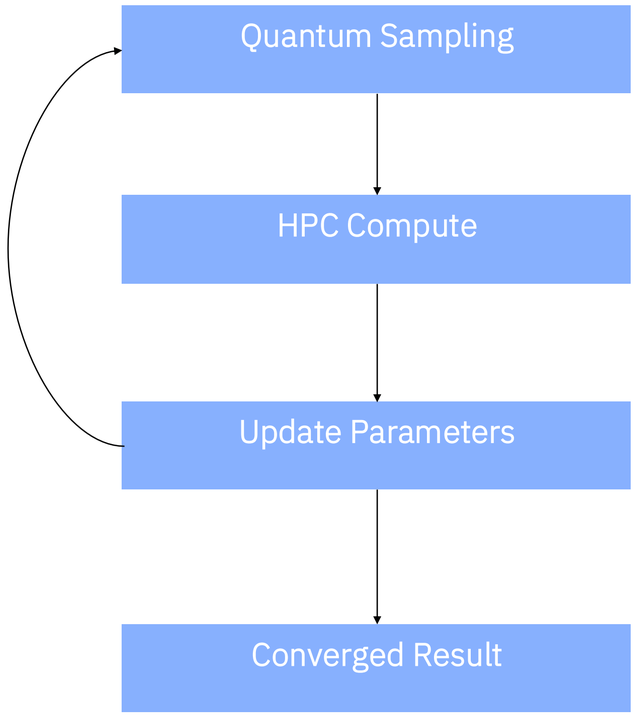
Our target workflow is not a one-shot job. It includes:

- quantum sampling
- classical post-processing
- HPC execution
- iterative parameter updates

With ad hoc scripts, it becomes hard to:

- track parameters and outputs
- recover from failures
- compare runs reproducibly
- manage the full loop across systems





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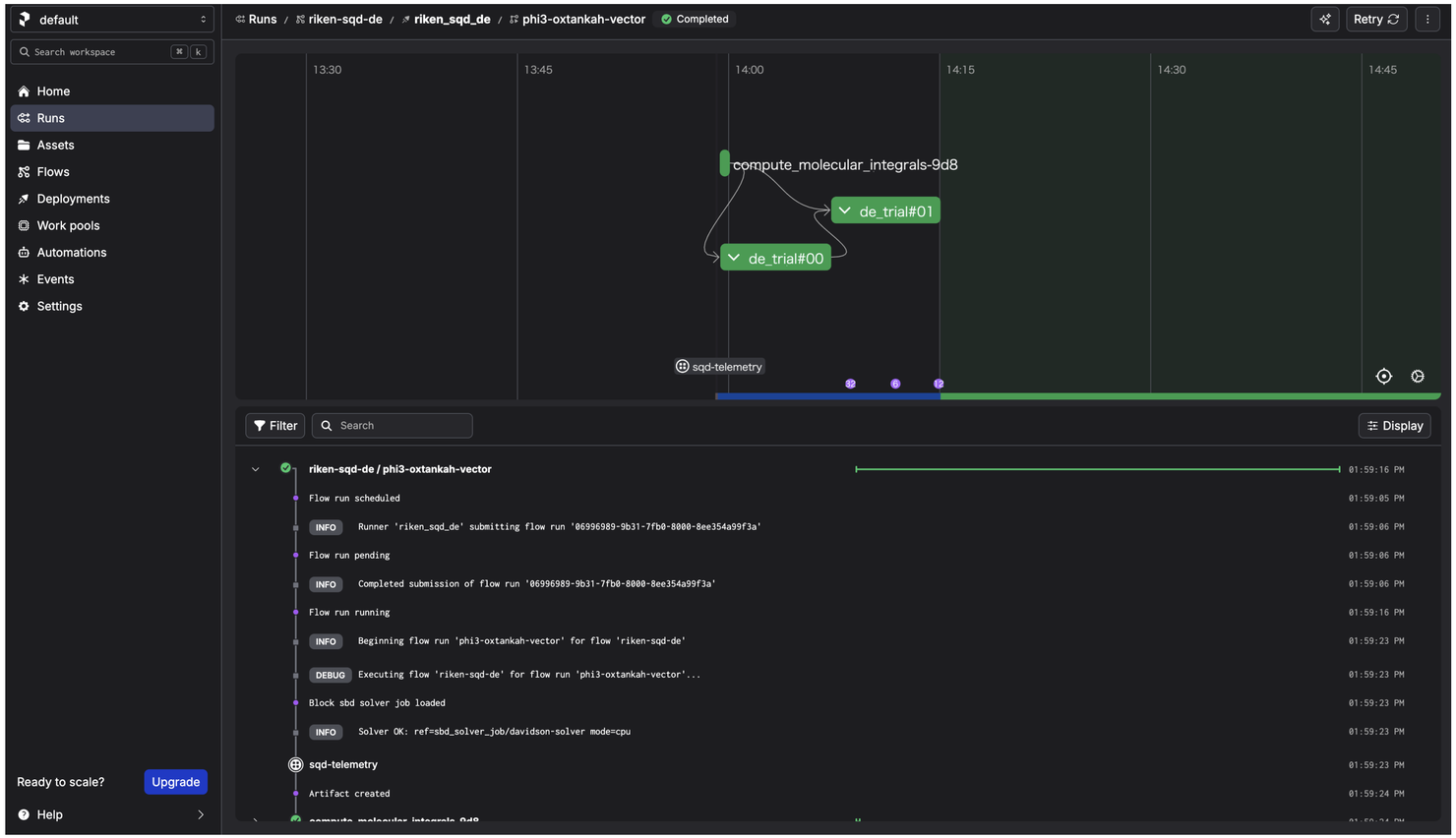
What Prefect Does

Prefect is a Python workflow orchestration framework.

It helps us:

- define the workflow in Python
- manage execution order
- track parameters, logs, and states
- rerun failed parts more clearly
- connect quantum and HPC steps in one workflow

Prefect orchestrates the workflow.
Fugaku executes the heavy computation.



Four Basic Ideas in Prefect

Flow (end-to-end workflow)

- The Flow represents the full QCSC iteration

Tasks (individual execution steps)

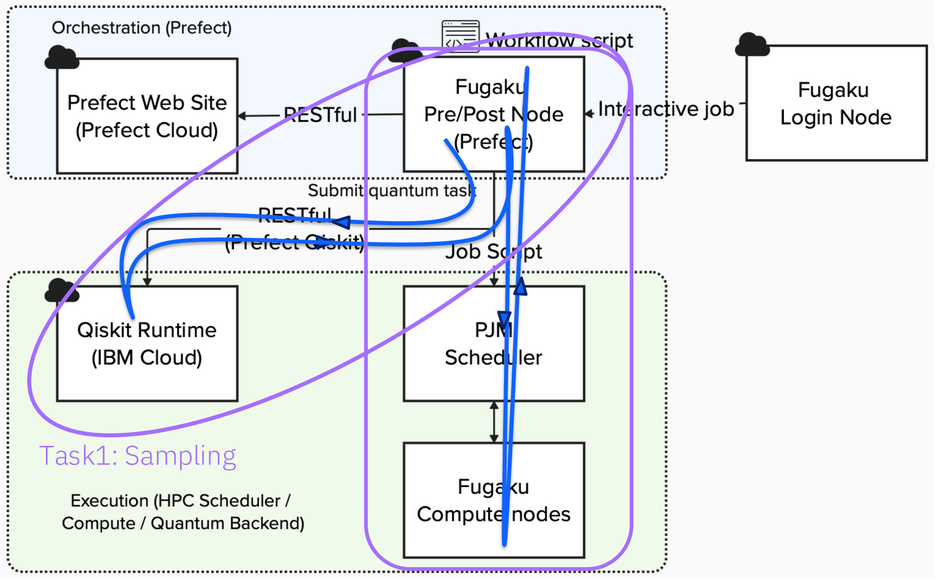
- Sampling Task: run the Sampler.
- HPC Task: submit a PJM job to run MPI programs.

Blocks (reusable configuration + credentials)

- Quantum Blocks: IBM Quantum credentials / runtime configuration.
- HPC Blocks: PJM job configuration.

Variables (run-time parameters)

- Quantum parameters: shots, circuit or sampler options.
- Algorithm parameters: iteration used by the workflow logic.



Task2: HPC Task

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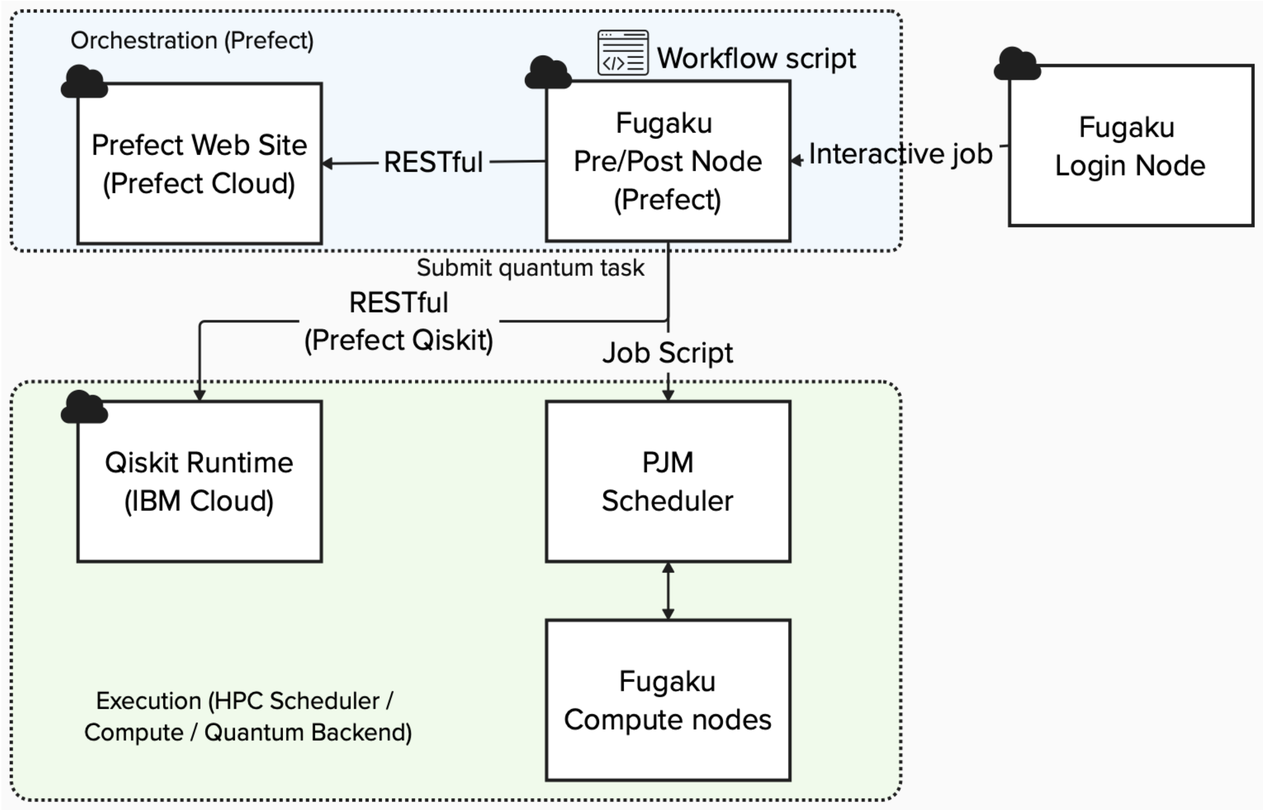
Where Things Run

Fugaku Login node
- used to build the SBD executable
- submits the interactive job for Pre/Post Node

Pre/Post Node
- runs the Prefect flow
- handles Python-side workflow logic
- submits HPC jobs
- does NOT run heavy computation

Qiskit Runtime
- executes quantum sampling

Fugaku Compute nodes
- run SBD via PJM/MPI



What We Prepare Once

One-time preparation:

On the Pre/Post node:

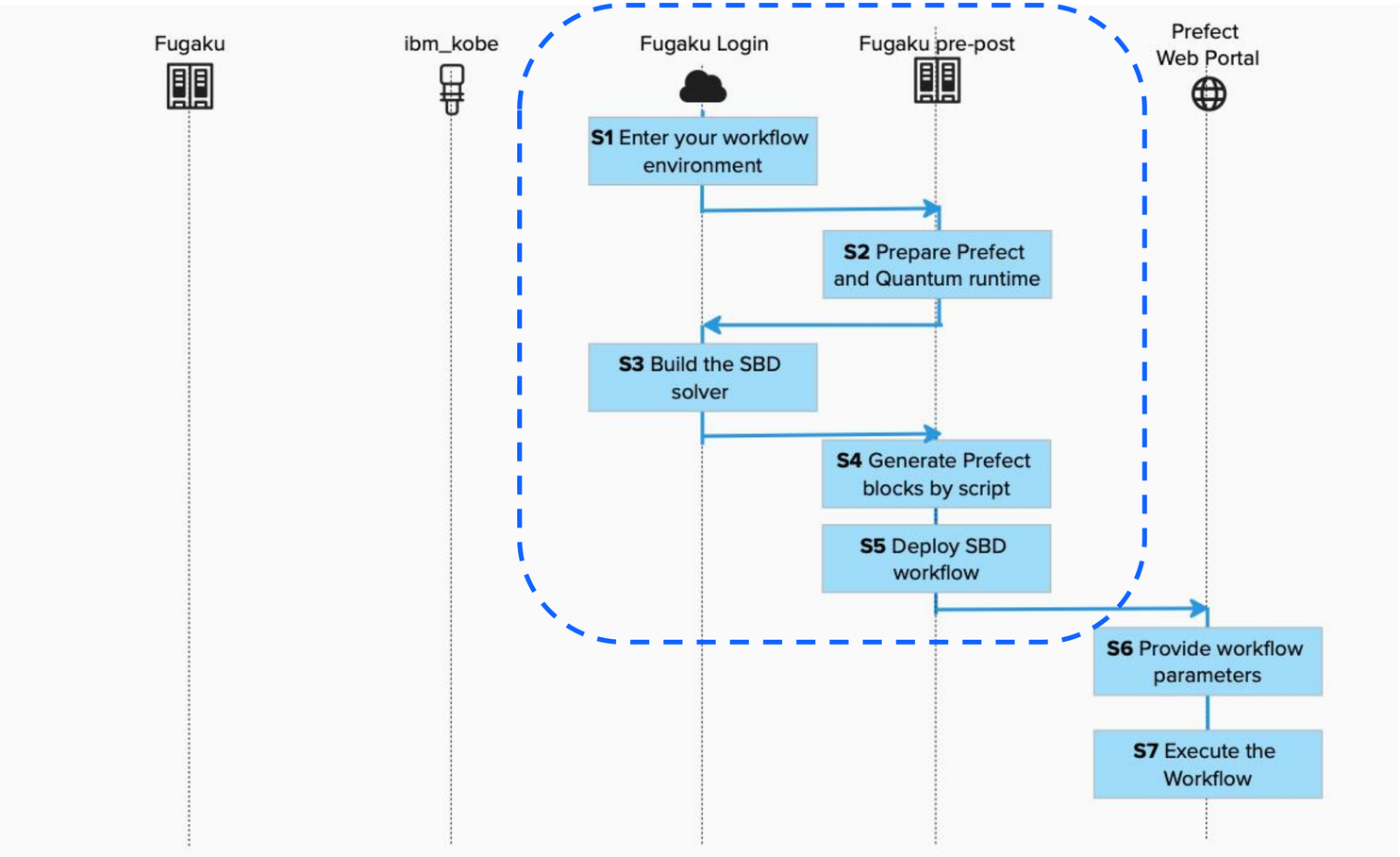
S2: Install Prefect and workflow packages and configure quantum access

S4: create saved execution settings in Prefect

S5: create the deployment

On the Login node:

S3: build the SBD executable (diag)



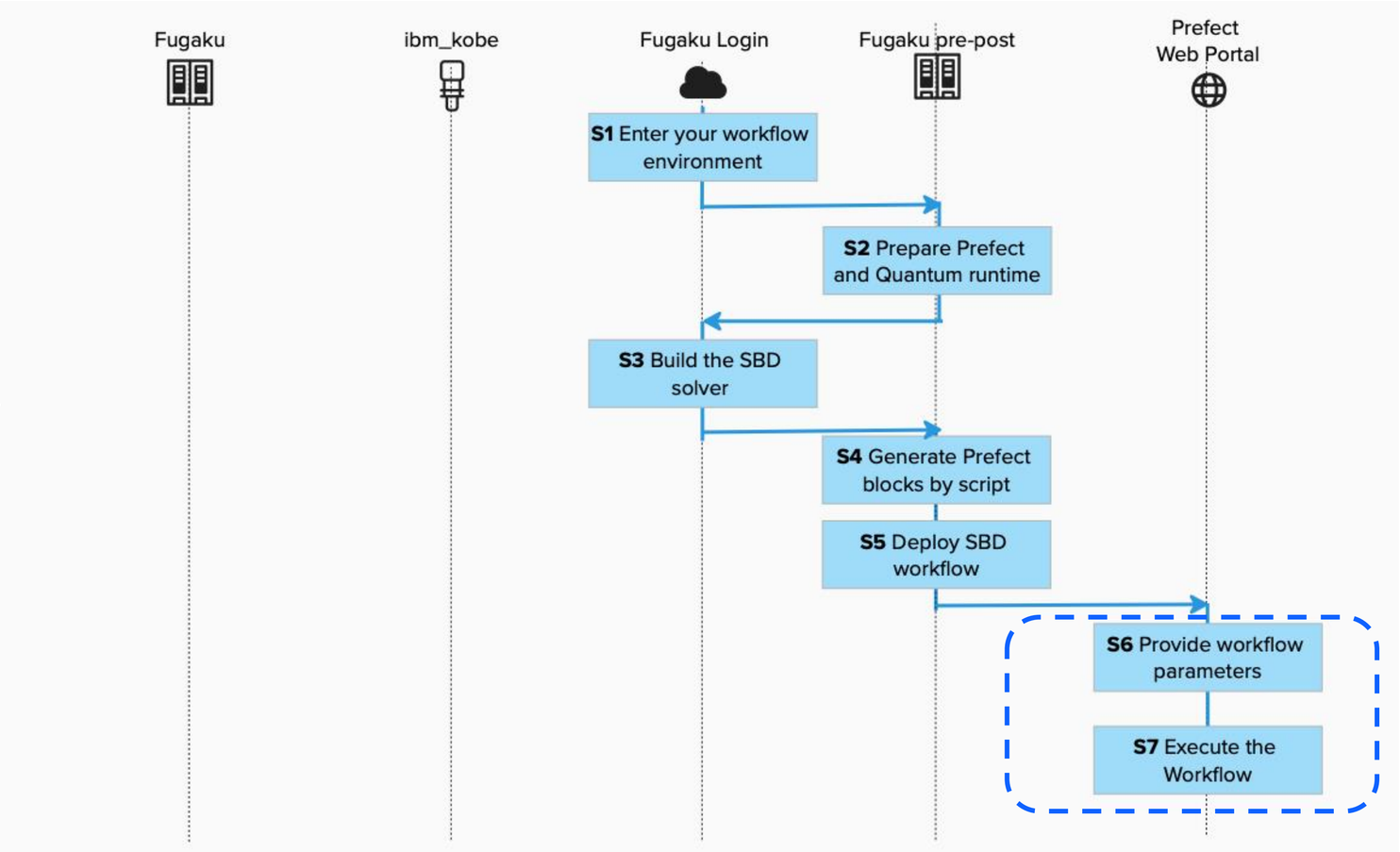
What We Do for Each Run

Per-run user actions

On the Pre/Post node:

S6: choose input data and set a few runtime parameters

S7: submit the workflow and monitor executior



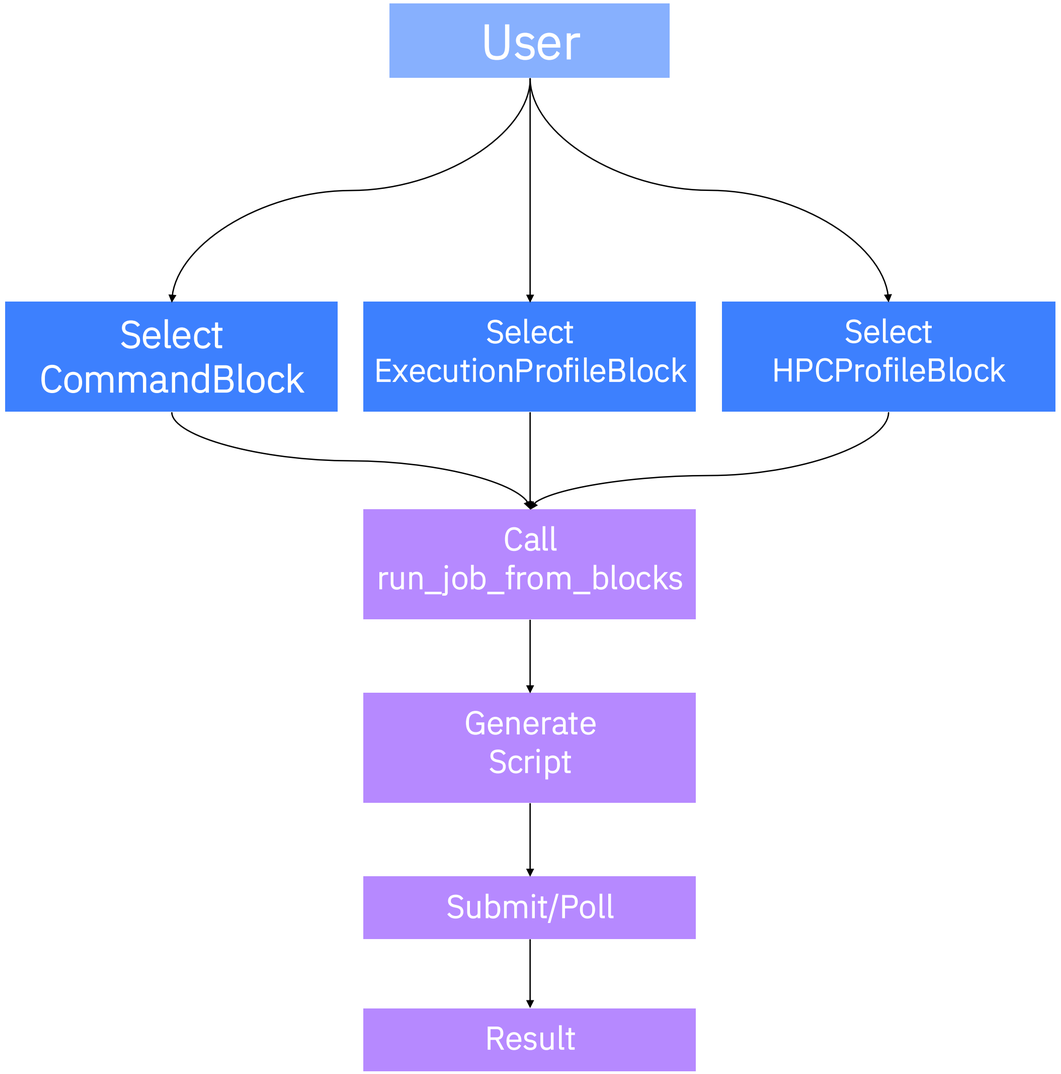
How Prefect + HPC-Prefect Simplifies Operations

- One API call triggers HPC execution
- Validation, script rendering, submit, and polling are automated
- Failures and retries are visible in Prefect

```
await run_job_from_blocks(
    command_block_name="cmd-diag",
    execution_profile_block_name="exec-diag-mpi",
    hpc_profile_block_name="hpc-miyabi",
    work_dir="./work",
    script_filename=”diag",
)
```

3-Step User Operation

1. **Select CommandBlock**: command identity (WHAT)
2. **Select ExecutionProfileBlock**: runtime shape (HOW)
3. **Select HPCProfileBlock**: target environment (WHERE)



Switch HPC by Changing 1-2 Lines

- Workflow logic stays unchanged.
- Change **hpc_profile_block_name**

```
await run_job_from_blocks(
    command_block_name="cmd-diag",
    execution_profile_block_name="exec-diag-mpi",
    hpc_profile_block_name="hpc-miyabi" or "hpc-fugaku",
    work_dir="./work",
    script_filename="diag",
)
```